

HANUAI - ML ASSESMENT

Background:

You have recently joined HanuAi, a leading technology company revolutionizing engineering quality and safety with AI-driven solutions. HanuAi's innovative tools leverage cutting-edge AI technologies, including Generative AI and Large Language Models (LLMs), to address complex challenges in data processing, automation, and analytics for Road and Infra Maintenance.

This task is designed to evaluate your data analytics and value addition approach to solve customer problems. The response of this assignment will be evaluated on –

- Understanding of data and tools used to perform the analysis
- Value generation via potential insights found during analysis
- Documentation/presentation to stakeholders

Task 1: Web Scraping and Sentiment Analysis

Objective - Write a Python script to scrape product reviews from an e-commerce website - www.bestbuy.ca/en-ca. The script should handle multiple filters, paginate through review pages, and extract key review details. The code should only extract the data available for public view and should not violate any website policy.

(Choose any product that has at least 50 reviews from the BestBuy Canada website)

Guidelines

1. **Setup** - Choose a Web Scraping Library: Use a Python web scraping library such as Selenium, BeautifulSoup, or similar tools.
 - a. Apply Filters and Load Pages: Implement functionality to apply multiple filters and load all pages to maximize the number of reviews mined.
2. **Data Scraping**- Extract Required Details having below fields, but not limited to:
 - a. Primary Key
 - b. Title
 - c. Review Text
 - d. Date (formatted as YYYY-MM-DD)
 - e. Rating (out of 5)
 - f. Source
 - g. Reviewer Name
3. **Additional comments on scraping**
 - a. Pagination and Filters - Handle Pagination by clicking the "Show More" buttons to access all available reviews.
 - b. Use filters such as "Most Helpful," "Newest," "Highest Rating," "Lowest Rating," and "Most Relevant" to refine the review extraction process.

4. **Sentiment Analysis** – Provide a code example that demonstrates the complete process loading the review text, preprocessing it if necessary, applying a suitable model or technique for sentiment analysis, and categorizing the results.
- a. Refer below image as an example of sentiment categorisation

ID	TITLE	REVIEW	SENTIMENT
R10876HKPBLSPQ	Very nice!	Colour is very pretty...easy to assemble! Bought for granddaughter!	['Good Design Quality (Pos)'] C
R1091K6DK0TN0D	Mmm	I like the is not to bulky and it works perfect for rear facing when front facing it dsnt rotate that why I'm giving it 4 stars	['Satisfactory Recommended (Pos)'] C

5. **Suggested solution for scraping challenges**

- a. Suggest ways to handle anti-scraping blockers e.g. rotating proxies, CAPTCHA bypass, or retry mechanisms, IP blocking.

6. **Business Context**

- a. Provide a summary of insights, such as top drivers of customer satisfaction and dissatisfaction, with recommendations for improvement.

Deliverables

1. Share the Python script (.ipynb) used for web scraping and sentiment analysis. (Include explanation of your code as comments in script.)
2. Submit the excel/csv file highlight outputs of scraped fields and sentiment categorisation.
3. Draft a report/doc including –
 - a. Solution for scraping challenges
 - b. Summary of potential insights for stakeholders, with recommendation for improvement

TASK 2: Advanced EDA and Text Mining

Objective – Perform comprehensive EDA, text mining, and actionable insight generation using the provided dataset.

Guidelines –

1. **Exploratory Data Analysis (Using python)**

- a. To understand data types, volume and characteristic per column. Identify and solve for missing values, duplicated and other inconsistencies found.

- b. Identify critical columns that looks insightful for stakeholders and provide a summary of key statistical analysis along with distribution/visualizations for critical columns identified.
2. Text Mining (Using NLP/open source LLM based)
 - a. Transform unstructured data into Structure format –
 - i. To identify free text columns given in raw data.
 - ii. Use text mining techniques to extract meaningful tags from the free text fields. *(Refer to few examples of entities (but not limited to) given in excel file)*
 - b. Categories “type of issues” (e.g. component failure, electrical issue etc.) based upon observation/grouping of mined entities.
3. Insights for stakeholders
 - a. Identify patterns and trends, highlighting significant changes in event frequencies, e.g. most occurring failure reported
 - b. Provide actionable recommendation for stakeholders to improve product quality and performance based upon your analysis.
 - c. Apply clustering or topic modelling techniques to group failure modes or customer complaints into categories.
 - i. identify potential root causes for recurring failures.

Deliverables

1. Submit the python script (.ipynb) used for analysis. *(Include a explanation of your code as comments in script.)*
2. Submit the excel/csv file highlight extracted tags in text mining.
3. Draft a report/doc including –
 - Overall synthesis of your understanding of the data, possible analysis outcomes and potential value generation for customers.
 - Summary of actionable points for stakeholders detailing business impact as per the insights generated from raw + extracted/mined data
 - Key learnings and further improvements on this task.